

Neural Architecture Search for RAPTOR Re-estimation: Direct-Rating and Player-vs-Player Ranking Models

nba-rapture

August 12, 2026

Abstract

We run two neural-architecture-search campaigns against the strongest gradient-boosted baselines in this project: (i) direct regression of per-player-season offensive and defensive RAPTOR, and (ii) a player-vs-player (pairwise) ranking formulation in which the model only predicts which of two players has the higher rating. Search spaces cover four regression families (plain MLP, residual MLP, self-normalizing SELU networks, bottleneck compressors) and three pairwise families (Bradley–Terry scorer, difference network, two-tower interaction encoder), with successive-halving over depth, width, activation, normalization, dropout, learning rate, weight decay, batch size, and schedule. Selection uses held-out whole-season validation cells; the 2013-14 and 2014-15 test cells are scored exactly once per final model. We report top-of-leaderboard metrics (dev@10, dev@20, Kendall $\tau@k$, hits@k), side-by-side projected-vs-true leaderboards including an in-sample training season, and comparisons against Neil Paine’s Estimated RAPTOR linear model.

1 Task, data, and splits

Labels. FiftyThreeEight full-history RAPTOR, separate offense (`raptor_offense`) and defense (`raptor_defense`) per player per season. **Features.** The project’s scraped matrix: pbstats totals/on-off, WOWY and opponent-WOWY splits, NBA tracking tables, and `leaguedashptdefend` nearest-defender data; after rate normalization, on-minus-off differentials, cell-relative standardization (12 features), and engineered opponent/defend blocks, offense rows carry 1,173 columns and defense rows 1,161. Identifier columns are excluded.

Splits (identical for every system in this report):

- *Train*: 13,021 regular-season rows (≥ 50 minutes) from labeled cells spanning 2015-16 through 2022-23, in-season snapshots included, excluding the four validation cells below.
- *Validation*: four whole-season cells (2015-16, 2016-17, 2017-18, 2019-20) used for NAS selection and early stopping only.
- *Test*: the 2013-14 and 2014-15 regular-season cells (494 rows), never touched during search; each final model is scored on them once.

Metrics. dev@ k = mean absolute rank error of the projected top- k (how far the player projected at position $i \leq k$ truly ranks from i); $\tau@k$ = Kendall tau between projected and true order of the true top- k ; hits@ k = overlap of projected and true top- k sets, summed over both test cells; MAE in RAPTOR points (undefined for tournament-scale pairwise systems).

2 Search methodology

Direct-rating NAS samples 24 configurations from: family $\in \{\text{mlp, resmlp, snn, bottleneck}\}$; width $\in \{128, 256, 512, 1024\}$; depth 1–4 (2–6 for resmlp); activation $\{\text{ReLU, GELU, SiLU}\}$; normalization $\{\text{batch, layer, none}\}$; dropout $\{0, 0.1, 0.2, 0.35\}$; lr $\log \mathcal{U}(10^{-3.7}, 10^{-2.3})$; weight decay $\{0, 10^{-5}, 10^{-4}, 10^{-3}\}$; batch $\{256, 512, 1024\}$; cosine schedule on/off. SNN ignores the sampled activation/norm and uses SELU with LeCun-normal initialization and alpha-dropout. Loss is smooth-L1 weighted by $\sqrt{\text{minutes}}$. Successive halving: 24×30 epochs $\rightarrow$ top $8 \times +60 \rightarrow$ top $3 \times +120$; selection on mean validation-cell Spearman; the winner is retrained from scratch with three seeds and seed-averaged.

Pairwise NAS (round 2) samples 20 configurations from families $\{\text{scorer } s(a) - s(b), \text{antisymmetrized difference net } f(a - b), \text{two-tower } g([e_a - e_b, e_a e_b])\}$, each with optional residual trunk, plus the same hyperparameter axes

and an embedding size $\{32, 64, 128\}$ for the two-tower. Training streams 200k fresh within-cell ordered pairs per epoch (ties $|\Delta y| < 0.05$ dropped), BCE on the pair logit. Halving: 20×8 epochs $\rightarrow 6 \times +16 \rightarrow 3 \times +24$; selection on validation-cell pairwise accuracy. Inference is a full round-robin tournament per cell with antisymmetrized win probabilities; we evaluate both the solo winner and the ensemble of the three best architectures (round 1 fielded only solo models while the GBM baseline enjoys seed averaging). Round 1 (20 configs, no residual trunks, no schedules, offense only) is included for reference.

3 Winning architectures

Table 1: NAS winners and their full hyperparameters. The SNN family overrides act/norm with SELU/none by construction.

param	direct offense	direct defense	pairwise offense	pairwise defense
family	snn	resmlp	diffnet	diffnet
res	–	–	False	False
width	128	512	256	256
depth	2	4	3	3
act	silu	gelu	gelu	gelu
norm	batch	none	layer	layer
dropout	0.35	0.2	0.2	0.2
lr	2.03e-03	2.35e-03	1.27e-03	1.27e-03
wd	0.0001	0.0001	1e-05	1e-05
batch	512	512	4096	4096
sched	none	cosine	cosine	cosine
emb	–	–	128	128
bneck	64	64	–	–
val score	+0.9092	+0.8398	0.8881	0.8322

The pairwise ensembles average the three best architectures per target: offense diffnet(0.8881), diffnet(0.8878), diffnet(0.8824); defense diffnet(0.8322), diffnet(0.8294), diffnet(0.8278). Pairwise round 1 winner (offense): scorer, hidden (256, 256), relu, norm=none, dropout 0.2, lr 6.89e-04, wd 0.0001, batch 4096; val acc 0.8832, test dev@10 3.60.

4 Results

Table 2: Offense, full test cells (2013-14 + 2014-15 RS, 494 rows). GBM systems are the project baselines; NAS systems are this report’s subject.

system	dev@10	dev@20	τ @10	τ @20	hits@10	hits@20	MAE
GBM direct blend	1.95	3.73	+0.600	+0.705	16/20	32/40	0.542
GBM pairwise (48k)	1.55	2.88	+0.756	+0.779	17/20	35/40	–
NAS direct net	3.65	6.12	+0.822	+0.632	14/20	30/40	0.750
NAS pairwise ens.	3.25	6.92	+0.711	+0.621	15/20	29/40	–

Table 3: Defense, full test cells (2013-14 + 2014-15 RS, 494 rows). GBM systems are the project baselines; NAS systems are this report’s subject.

system	dev@10	dev@20	τ @10	τ @20	hits@10	hits@20	MAE
GBM direct blend	3.65	6.25	+0.511	+0.532	14/20	29/40	0.684
GBM pairwise (48k)	4.65	7.12	+0.511	+0.453	14/20	31/40	–
NAS direct net	12.75	15.07	+0.311	+0.379	12/20	23/40	0.845
NAS pairwise ens.	6.00	12.38	+0.511	+0.442	12/20	27/40	–

For production context (different architecture, same test cells): the deployed offense stack (components + opponent features) verifies at dev@10 1.10, τ @10 +0.800; the deployed defense stack (matched features + defend) at dev@10 3.80, τ @10 +0.511. Ten-fold leave-one-season-out medians: offense 1.50, defense 5.00, pairwise GBM 1.60.

4.1 Against Neil Paine’s Estimated RAPTOR

Caveat favoring Paine: his published weights were fit on full RAPTOR 2014–2023, which *contains both of our test seasons* – his numbers are in-sample where ours are strictly out-of-sample. Pools below are the ≥ 1065 -minute test rows matched into his file by normalized name (offense n=490, defense n=490).

Table 4: Offense, Paine-matched ≥ 1065 -minute pool.

system	dev@10	dev@20	τ @10	τ @20	hits@10	hits@20	MAE
GBM direct blend	1.95	3.77	+0.600	+0.695	16/20	33/40	0.544
GBM pairwise (48k)	1.55	2.92	+0.756	+0.789	17/20	34/40	–
NAS direct net	3.65	6.88	+0.822	+0.658	14/20	29/40	0.749
NAS pairwise ens.	3.25	7.25	+0.711	+0.605	15/20	29/40	–
Paine eRT (publ.)	4.35	6.38	+0.667	+0.663	17/20	32/40	0.718
Paine eRT (recr.)	4.35	9.35	+0.556	+0.679	16/20	31/40	0.784

Table 5: Defense, Paine-matched ≥ 1065 -minute pool.

system	dev@10	dev@20	τ @10	τ @20	hits@10	hits@20	MAE
GBM direct blend	3.80	6.30	+0.489	+0.505	14/20	29/40	0.680
GBM pairwise (48k)	4.75	6.97	+0.489	+0.453	14/20	31/40	–
NAS direct net	13.00	15.18	+0.289	+0.363	12/20	23/40	0.840
NAS pairwise ens.	6.05	12.40	+0.489	+0.416	12/20	27/40	–
Paine eRT (publ.)	13.40	16.85	+0.511	+0.432	11/20	26/40	0.885
Paine eRT (recr.)	16.40	19.20	+0.489	+0.400	11/20	22/40	0.997

4.2 In-sample fit (training cell)

The 2021-22 regular-season cell was part of every system’s training data; this table shows memorization capacity, not skill, and is included because projected-vs-true boards on a training season (next section) are only interpretable alongside it.

Table 6: Offense, 2021-22 RS, in-sample.

system	dev@10	dev@20	τ @10	τ @20	hits@10	hits@20	MAE
GBM direct blend	1.40	2.40	+0.689	+0.747	10/20	18/40	0.381
GBM pairwise (48k)	1.80	2.65	+0.733	+0.705	8/20	19/40	–
NAS direct net	0.40	0.55	+0.911	+0.937	10/20	19/40	1.067
NAS pairwise ens.	0.80	1.85	+0.778	+0.800	9/20	19/40	–

5 Leaderboards: projected vs. true

Each cell shows the player a system projects at that position, with his true rank in parentheses; pools are the ≥ 1065 -minute players of the cell. 2013-14 and 2014-15 are out-of-sample for our systems (in-sample for Paine); 2021-22 is in-sample for our systems and shown to make training-fit visible.

Table 7: Defense, 2021-22 RS, in-sample.

system	dev@10	dev@20	τ @10	τ @20	hits@10	hits@20	MAE
GBM direct blend	3.10	6.70	+0.556	+0.590	7/20	16/40	0.548
GBM pairwise (48k)	6.60	7.05	+0.467	+0.632	7/20	16/40	–
NAS direct net	0.10	0.20	+1.000	+0.979	9/20	20/40	0.005
NAS pairwise ens.	1.10	2.15	+0.689	+0.758	8/20	17/40	–

Table 8: Offense 2013-14 (test)

#	true (score)	GBM direct blend	GBM pairwise	NAS direct net	NAS pairwise ens.	Paine eRT
1	K. Durant (+7.6)	L. James (4)	K. Durant (1)	K. Durant (1)	K. Durant (1)	K. Durant (1)
2	C. Paul (+7.1)	K. Durant (1)	C. Paul (2)	C. Paul (2)	L. James (4)	L. James (4)
3	J. Harden (+6.3)	J. Harden (3)	L. James (4)	J. Harden (3)	J. Harden (3)	C. Paul (2)
4	L. James (+5.8)	C. Paul (2)	J. Harden (3)	L. James (4)	C. Paul (2)	J. Harden (3)
5	K. Love (+5.7)	G. Dragic (6)	K. Love (5)	K. Love (5)	K. Love (5)	G. Dragic (6)
6	G. Dragic (+4.8)	K. Love (5)	G. Dragic (6)	D. Lillard (12)	G. Dragic (6)	K. Love (5)
7	K. Lowry (+4.4)	D. Lillard (12)	K. Lowry (7)	G. Dragic (6)	D. Lillard (12)	D. Nowitzki (8)
8	D. Nowitzki (+4.4)	M. Ginobili (10)	D. Lillard (12)	I. Thomas (14)	K. Lowry (7)	B. Wright (41)
9	C. Anthony (+4.2)	R. Westbrook (15)	M. Ginobili (10)	K. Lowry (7)	P. George (22)	C. Anthony (9)
10	M. Ginobili (+4.0)	C. Anthony (9)	C. Anthony (9)	P. George (22)	I. Thomas (14)	K. Lowry (7)
11	P. Mills (+3.9)	K. Lowry (7)	D. Nowitzki (8)	C. Anthony (9)	M. Conley (13)	B. Griffin (17)
12	D. Lillard (+3.6)	I. Thomas (14)	M. Conley (13)	B. Griffin (17)	C. Anthony (9)	I. Thomas (14)
13	M. Conley (+3.5)	D. Nowitzki (8)	I. Thomas (14)	M. Ginobili (10)	M. Ginobili (10)	M. Ginobili (10)
14	I. Thomas (+3.5)	M. Conley (13)	R. Westbrook (15)	D. Nowitzki (8)	D. Nowitzki (8)	R. Westbrook (15)
15	R. Westbrook (+3.3)	P. Mills (11)	B. Griffin (17)	R. Westbrook (15)	B. Griffin (17)	N. Pekovic (68)

Table 9: Offense 2014-15 (test)

#	true (score)	GBM direct blend	GBM pairwise	NAS direct net	NAS pairwise ens.	Paine eRT
1	C. Paul (+8.5)	J. Harden (2)	C. Paul (1)	C. Paul (1)	J. Harden (2)	C. Paul (1)
2	J. Harden (+7.7)	C. Paul (1)	J. Harden (2)	J. Harden (2)	C. Paul (1)	J. Harden (2)
3	R. Westbrook (+6.1)	L. James (5)	L. James (5)	L. James (5)	K. Irving (4)	L. James (5)
4	K. Irving (+5.5)	R. Westbrook (3)	R. Westbrook (3)	D. Lillard (11)	L. James (5)	A. Davis (9)
5	L. James (+5.3)	K. Irving (4)	K. Irving (4)	R. Westbrook (3)	L. Williams (6)	R. Westbrook (3)
6	L. Williams (+5.2)	I. Thomas (8)	L. Williams (6)	L. Williams (6)	D. Lillard (11)	J. Butler (20)
7	K. Korver (+4.6)	D. Lillard (11)	D. Lillard (11)	K. Irving (4)	R. Westbrook (3)	B. Griffin (22)
8	I. Thomas (+4.5)	L. Williams (6)	K. Thompson (10)	I. Thomas (8)	K. Thompson (10)	K. Irving (4)
9	A. Davis (+4.3)	K. Thompson (10)	I. Thomas (8)	B. Griffin (22)	J. Redick (29)	L. Williams (6)
10	K. Thompson (+4.3)	G. Hill (12)	B. Griffin (22)	J. Redick (29)	I. Thomas (8)	K. Thompson (10)
11	D. Lillard (+4.0)	B. Griffin (22)	J. Redick (29)	K. Thompson (10)	J. Butler (20)	G. Hill (12)
12	G. Hill (+3.9)	A. Davis (9)	A. Davis (9)	K. Lowry (18)	K. Lowry (18)	K. Leonard (15)
13	C. Anthony (+3.8)	J. Butler (20)	G. Hill (12)	K. Leonard (15)	K. Korver (7)	J. Redick (29)
14	T. Lawson (+3.8)	M. Conley (32)	K. Korver (7)	B. Jennings (23)	B. Griffin (22)	T. Lawson (14)
15	K. Leonard (+3.7)	C. Anthony (13)	J. Butler (20)	K. Korver (7)	A. Davis (9)	G. Hayward (21)

Table 10: Defense 2013-14 (test)

#	true (score)	GBM direct blend	GBM pairwise	NAS direct net	NAS pairwise ens.	Paine eRT
1	K. Leonard (+5.0)	A. Bogut (4)	D. Green (2)	D. Green (2)	A. Bogut (4)	A. Bogut (4)
2	D. Green (+4.6)	T. Splitter (6)	A. Bogut (4)	R. Hibbert (15)	D. Green (2)	K. Leonard (1)
3	J. Noah (+4.5)	D. Green (2)	J. Noah (3)	A. Bogut (4)	J. Noah (3)	D. Green (2)
4	A. Bogut (+4.4)	K. Leonard (1)	T. Splitter (6)	Nene (9)	K. Leonard (1)	D. Jordan (65)
5	M. KiddGilchrist (+4.4)	P. George (21)	P. George (21)	A. Davis (34)	J. Butler (14)	J. Butler (14)
6	T. Splitter (+4.2)	K. Garnett (11)	K. Leonard (1)	D. Cousins (16)	Nene (9)	T. Allen (31)
7	D. Green (+4.0)	J. Noah (3)	Nene (9)	M. Gasol (22)	P. George (21)	J. Noah (3)
8	C. Paul (+3.9)	Nene (9)	R. Hibbert (15)	J. Noah (3)	R. Hibbert (15)	D. Green (7)
9	Nene (+3.8)	I. Mahinmi (13)	A. Iguodala (25)	T. Duncan (18)	A. Iguodala (25)	R. Rubio (49)
10	A. Varejao (+3.6)	J. Butler (14)	J. Butler (14)	D. Jordan (65)	D. Cousins (16)	A. Iguodala (25)
11	K. Garnett (+3.5)	D. Green (7)	I. Mahinmi (13)	A. Varejao (10)	A. Jefferson (47)	P. George (21)
12	N. Calathes (+3.5)	A. Varejao (10)	M. Gortat (29)	K. Garnett (11)	A. Varejao (10)	K. OQuinn (53)
13	I. Mahinmi (+3.5)	A. Iguodala (25)	D. Cousins (16)	P. George (21)	K. Hinrich (42)	R. Hibbert (15)
14	J. Butler (+3.4)	A. Davis (34)	K. Garnett (11)	T. Splitter (6)	D. Jordan (65)	D. West (57)
15	R. Hibbert (+3.4)	T. Allen (31)	C. Watson (17)	K. Leonard (1)	K. Garnett (11)	P. Millsap (23)

Table 11: Defense 2014-15 (test)

#	true (score)	GBM direct blend	GBM pairwise	NAS direct net	NAS pairwise ens.	Paine eRT
1	K. Leonard (+5.2)	D. Green (2)	D. Green (2)	D. Green (2)	D. Green (2)	K. Leonard (1)
2	D. Green (+5.1)	T. Allen (4)	A. Bogut (5)	A. Davis (6)	R. Gobert (3)	T. Allen (4)
3	R. Gobert (+4.8)	A. Bogut (5)	T. Allen (4)	R. Gobert (3)	A. Davis (6)	D. Green (2)
4	T. Allen (+4.8)	R. Gobert (3)	R. Gobert (3)	A. Bogut (5)	T. Allen (4)	A. Bogut (5)
5	A. Bogut (+4.7)	K. Leonard (1)	D. Cousins (7)	D. Cousins (7)	K. Leonard (1)	D. Jordan (63)
6	A. Davis (+4.5)	A. Davis (6)	N. Noel (19)	T. Duncan (9)	D. Cousins (7)	N. Noel (19)
7	D. Cousins (+4.4)	Z. Pachulia (12)	Z. Pachulia (12)	K. Leonard (1)	A. Bogut (5)	A. Davis (6)
8	M. Gortat (+3.6)	N. Noel (19)	M. Gortat (8)	D. Jordan (63)	M. Smart (43)	R. Gobert (3)
9	T. Duncan (+3.5)	D. Cousins (7)	K. Leonard (1)	M. Smart (43)	N. Noel (19)	D. Green (16)
10	A. Roberson (+3.4)	A. Roberson (10)	A. Davis (6)	T. Allen (4)	Z. Pachulia (12)	A. Aminu (24)
11	K. Koufos (+3.3)	K. Koufos (11)	D. Green (16)	P. Gasol (90)	E. Ilyasova (100)	T. Duncan (9)
12	Z. Pachulia (+3.2)	J. Smith (23)	A. Aminu (24)	Z. Pachulia (12)	Nene (18)	P. Millsap (26)
13	K. Middleton (+3.1)	Nene (18)	M. KiddGilchrist (14)	D. Favors (34)	A. Aminu (24)	K. Middleton (13)
14	M. KiddGilchrist (+3.0)	M. Gortat (8)	Nene (18)	A. Aminu (24)	A. Iguodala (47)	A. Roberson (10)
15	S. Ibaka (+3.0)	D. Green (16)	K. Koufos (11)	M. Gasol (21)	M. KiddGilchrist (14)	B. Biyombo (109)

Table 12: Offense 2021-22 (training season, in-sample)

#	true (score)	GBM direct blend	GBM pairwise	NAS direct net	NAS pairwise ens.
1	N. Jokic (+9.0)	N. Jokic (1)	T. Young (2)	N. Jokic (1)	T. Young (2)
2	T. Young (+7.6)	T. Young (2)	N. Jokic (1)	T. Young (2)	N. Jokic (1)
3	L. Doncic (+5.8)	S. Curry (5)	L. Doncic (3)	L. Doncic (3)	L. Doncic (3)
4	G. Antetokounmpo (+5.7)	L. Doncic (3)	S. Curry (5)	S. Curry (5)	G. Antetokounmpo (4)
5	S. Curry (+5.7)	D. Mitchell (8)	G. Antetokounmpo (4)	G. Antetokounmpo (4)	S. Curry (5)
6	K. Durant (+5.0)	G. Antetokounmpo (4)	D. Mitchell (8)	K. Durant (6)	D. Mitchell (8)
7	J. Tatum (+4.8)	K. Durant (6)	K. Durant (6)	J. Tatum (7)	J. Tatum (7)
8	D. Mitchell (+4.6)	L. James (9)	J. Holiday (15)	L. James (9)	K. Durant (6)
9	L. James (+4.6)	J. Morant (10)	J. Morant (10)	D. Mitchell (8)	J. Morant (10)
10	J. Morant (+4.6)	J. Tatum (7)	D. Booker (13)	J. Morant (10)	D. Garland (11)
11	D. Garland (+4.5)	D. Garland (11)	J. Tatum (7)	D. Garland (11)	D. Booker (13)
12	J. Embiid (+4.3)	D. DeRozan (16)	D. Garland (11)	D. Booker (13)	L. James (9)
13	D. Booker (+4.3)	D. Booker (13)	L. James (9)	C. Paul (14)	J. Embiid (12)
14	C. Paul (+4.1)	J. Holiday (15)	J. Harden (18)	J. Embiid (12)	J. Harden (18)
15	J. Holiday (+4.0)	K. Towns (19)	J. Embiid (12)	J. Holiday (15)	D. DeRozan (16)

Table 13: Defense 2021-22 (training season, in-sample)

#	true (score)	GBM direct blend	GBM pairwise	NAS direct net	NAS pairwise ens.
1	R. Gobert (+6.9)	R. Gobert (1)	R. Gobert (1)	R. Gobert (1)	R. Gobert (1)
2	N. Jokic (+5.9)	N. Jokic (2)	M. Thybulle (6)	N. Jokic (2)	N. Jokic (2)
3	A. Caruso (+4.7)	J. Embiid (12)	J. Allen (7)	A. Caruso (3)	M. Thybulle (6)
4	D. Green (+4.3)	M. Thybulle (6)	D. Green (4)	D. Green (4)	A. Caruso (3)
5	G. Payton II (+4.3)	J. Allen (7)	N. Jokic (2)	G. Payton II (5)	G. Payton II (5)
6	M. Thybulle (+4.2)	G. Payton II (5)	A. Caruso (3)	M. Thybulle (6)	J. Allen (7)
7	J. Allen (+4.2)	D. Green (4)	K. Williams (11)	J. Allen (7)	P. George (8)
8	P. George (+3.9)	J. Nurkic (13)	G. Payton II (5)	P. George (8)	J. Embiid (12)
9	B. Adebayo (+3.9)	K. Williams (11)	I. Hartenstein (52)	B. Adebayo (9)	B. Adebayo (9)
10	A. Horford (+3.8)	A. Caruso (3)	J. Embiid (12)	K. Williams (11)	K. Williams (11)
11	K. Williams (+3.8)	I. Hartenstein (52)	B. Adebayo (9)	A. Horford (10)	D. Green (4)
12	J. Embiid (+3.6)	P. George (8)	A. Horford (10)	J. Embiid (12)	J. Nurkic (13)
13	J. Nurkic (+3.4)	R. Williams III (21)	R. Covington (20)	J. Nurkic (13)	A. Horford (10)
14	D. White (+3.2)	A. Horford (10)	M. Kleber (16)	D. White (14)	M. Kleber (16)
15	J. Smith (+2.9)	G. Antetokounmpo (36)	R. Williams III (21)	J. Smith (15)	D. White (14)

6 Discussion

- **Round 2 genuinely improved the neural pairwise model.** Adding difference-network depth, LayerNorm, and cosine schedules to the search moved the best neural pairwise result from dev@10 3.60 (round 1) to 2.70 on offense, and from 18.1 (the original all-pairs RankNet) to 6.05 on defense. Notably, the same configuration won both targets under independent selection, and its family is the free-form difference net $f(x_a - x_b)$ – the GBM’s own formulation – over the Bradley–Terry scorer that round 1’s halving kept.
- **But the GBM gap remains.** The best neural systems reach 2.70/6.05 (offense/defense) against 1.55/3.65 for GBMs consuming identical features and 1.10/3.80 for the production stacks. Across 108 trained configurations in five campaigns, no searched region closes it.
- **The gap is generalization, not capacity.** In-sample on the 2021-22 training cell the nets essentially memorize – direct net dev@10 0.40 offense, 0.10 defense (tau@10 +1.00) – while the GBMs fit training far more loosely (1.40/3.10) yet generalize better. On 13k rows $\times$ 1.1k columns the trees’ inductive bias, not model capacity, is what transfers.
- **Ensembling did not help.** Averaging the top-3 architectures moved offense from 2.70 to 3.25 and defense from 6.05 to 6.00: diverse architectures at a shared plateau average toward the plateau, unlike GBM seed-averaging, which reduces the variance of one stronger learner.
- **Defense stays qualitatively harder for networks** (best net 6.05 vs. 3.65–3.80 GBM), consistent with the elite-defender study: defensive elite ordering rides on a few engineered nearest-defender columns inside a full profile – a regime trees exploit and MLPs dilute.
- **Against Paine’s Estimated RAPTOR** every system here – including the neural ones – clears his published defense numbers (13.40 dev@10), and all GBM systems clear his offense (4.35), despite his weights being fit on data containing our test seasons.

- **Conclusion.** Gradient boosting remains the production choice for both formulations. Untested neural territory that plausibly differs in kind: attention over feature tokens (FT-Transformer) and prior-fitted models (TabPFN), both wanting GPU-scale compute.

Reproduction

```
python training/nas_direct.py; python training/nas_pairwise2.py; python training/build_nas_report_data.py;  
python training/gen_nas_report.py.
```